EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 8A

1. Basic DT-LTI Systems

A discrete-time system maps an input sequence $x[n]$ to an output sequence $y[n]$.

Consider the system H described by

$$y[n] = 3x[n] - 2x[n-1] + x[n-2] \qquad \text{for all } n \in \mathbb{Z}.$$

(a) Prove that H is linear.

Answer: Let $x_1[n] \to y_1[n]$ and $x_2[n] \to y_2[n]$, so that

$$y_1[n] = 3x_1[n] - 2x_1[n-1] + x_1[n-2], \qquad y_2[n] = 3x_2[n] - 2x_2[n-1] + x_2[n-2].$$

Now feed in the input $x_3[n] = \alpha x_1[n] + \beta x_2[n]$. The output is

$$\begin{aligned} y_3[n] &= 3x_3[n] - 2x_3[n-1] + x_3[n-2] \\ &= 3\left(\alpha x_1[n] + \beta x_2[n]\right) - 2\left(\alpha x_1[n-1] + \beta x_2[n-1]\right) + \left(\alpha x_1[n-2] + \beta x_2[n-2]\right) \\ &= \alpha\left(3x_1[n] - 2x_1[n-1] + x_1[n-2]\right) + \beta\left(3x_2[n] - 2x_2[n-1] + x_2[n-2]\right) \\ &= \alpha y_1[n] + \beta y_2[n]. \end{aligned}$$

Since this holds for every pair of inputs and every α, β, the system is linear.

(b) Prove that H is time invariant.

Answer: Let $x[n] \to y[n]$, and define the shifted input $x_2[n] = x[n-k]$ for some integer k. Plugging x_2 into the system,

$$\begin{aligned} y_2[n] &= 3x_2[n] - 2x_2[n-1] + x_2[n-2] \\ &= 3x[n-k] - 2x[(n-1)-k] + x[(n-2)-k] \\ &= 3x[n-k] - 2x[(n-k)-1] + x[(n-k)-2] \\ &= y[n-k], \end{aligned}$$

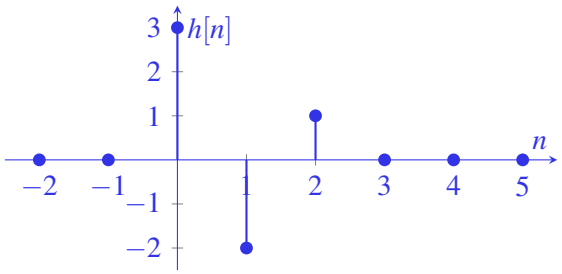
where the last line is the original equation evaluated at time $n-k$. Shifting the input by k shifts the output by k, so the system is time invariant.

(c) Find the impulse response $h[n]$ of H, that is, the output when the input is $x[n] = \delta[n]$. Sketch $h[n]$.

Answer: Plugging in $x[n] = \delta[n]$,

$$h[n] = 3\delta[n] - 2\delta[n-1] + \delta[n-2] = \begin{cases} 3, & n = 0, \\ -2, & n = 1, \\ 1, & n = 2, \\ 0, & \text{otherwise.} \end{cases}$$

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Notice that the impulse response just lists the coefficients of the difference equation.

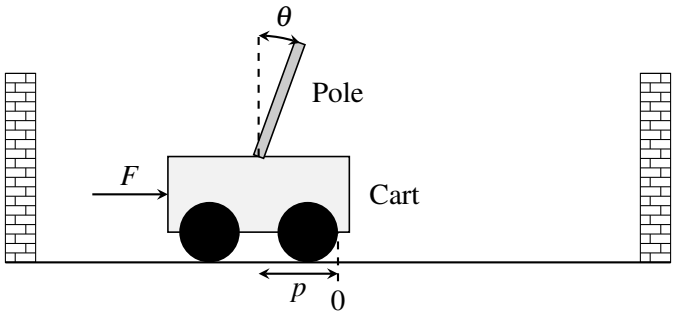
2. Segway Tours

Your friend has decided to start a new SF tour business, and you suggest they use segways.

They become intrigued by your idea and asks you how a segway works. A segway is essentially a stand on two wheels.

The segway works by applying a force (through the spinning wheels) to the base of the segway. This controls both the position on the segway and the angle of the stand. As the driver pushes on the stand, the segway tries to bring itself back to the upright position, and it can only do this by moving the base.

Is it possible for the segway to be brought upright and to a stop from any initial configuration? There is only one input (force) used to control two outputs (position and angle). You both talk to a third friend, who is a TA for EE 128, and she tells you that a segway can be modeled as a cart-pole system.



A cart-pole system can be fully described by its position p, velocity $\dot{p}$, angle θ, and angular velocity $\dot{\theta}$. We write this as a "state vector", $\mathbf{x}$:

$$\mathbf{x} = \begin{bmatrix} p \\ \dot{p} \\ \theta \\ \dot{\theta} \end{bmatrix}.$$

The input to this system is a scalar quantity $u[n]$ at time n, which is the force F applied to the cart (or base of the segway).

The cart-pole system can be represented by a linear model:

$$\mathbf{x}[n+1] = \mathbf{A}\mathbf{x}[n] + \mathbf{b}\,u[n], \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{4\times 4}$ and $\mathbf{b} \in \mathbb{R}^{4\times 1}$.

The control $u[n]$ allows us to move the state ($\mathbf{x}$) in the direction of $\mathbf{b}$. So, if $u[n] = 2$, we move the state by $2\mathbf{b}$ at time n, and so on. We can choose different controls at different times.

The model tells us how the the state vector, $\mathbf{x}$, will evolve over time as a function of the current state vector and control inputs.

You look at this general linear system and try to answer the following question: Starting from some initial state $\mathbf{x}_0$, can we reach a final desired state, $\mathbf{x}_f$, in N steps?

The challenge seems to be that the state is four-dimensional and keeps evolving and that we can only apply a one-dimensional (scalar) control at each time. Typically, to set the values of four variables to desired quantities, you would need four inputs. Can you do this with just one input?

We will solve this problem by walking through several steps.

(a) Before we start computing, let us think about what the control inputs let us do. Recall the system we have:

$$\mathbf{x}[n+1] = \mathbf{A}\mathbf{x}[n] + \mathbf{b}\,u[n].$$

The control allows us to move the state at time step $n+1$ by $u[n]$ in direction $\mathbf{b}$, remember $u[n] \in \mathbb{R}$ is just a scalar. Applying the model twice (you will derive this in part (c)) gives

$$\mathbf{x}[2] = \mathbf{A}^2\mathbf{x}[0] + \mathbf{A}\mathbf{b}\,u[0] + \mathbf{b}\,u[1].$$

Again, here $u[0], u[1] \in \mathbb{R}$ can be thought of as arbitrary scalars, and $\mathbf{A}\mathbf{b}\,u[0] + \mathbf{b}\,u[1]$ can be thought of as the set of all linear combinations of the vectors $\mathbf{b}$ and $\mathbf{A}\mathbf{b}$. Using this observation, can you express the possible states you can arrive at in two time steps using the span of exactly *two* vectors plus a vector offset?

Answer: You can move in directions $\mathbf{b}$ and $\mathbf{A}\mathbf{b}$. We also need to account for the zero input behavior of the system reflected by the term $\mathbf{A}^2\mathbf{x}[0]$. Hence you can reach all positions that are in

$$\text{span}\,\{\mathbf{b}, \mathbf{A}\mathbf{b}\} + \mathbf{A}^2\mathbf{x}[0].$$

(b) Express $\mathbf{x}[1]$ in terms of $\mathbf{x}[0]$ and the input $u[0]$. (*Hint:* This is not meant to be complicated and your answer may contain other relevant variables such as $\mathbf{A}$, $\mathbf{b}$.)

Answer:

From Equation (1), we get (by substituting $n = 0$):

$$\mathbf{x}[1] = \mathbf{A}\mathbf{x}[0] + \mathbf{b}\,u[0] \tag{2}$$

(c) i. Express $\mathbf{x}[2]$ in terms of *only* $\mathbf{x}[0]$ and the inputs, $u[0]$ and $u[1]$.

ii. Then express $\mathbf{x}[3]$ in terms of *only* $\mathbf{x}[0]$ and the inputs, $u[0]$, $u[1]$, and $u[2]$.

iii. Finally express $\mathbf{x}[4]$ in terms of *only* $\mathbf{x}[0]$ and the inputs, $u[0]$, $u[1]$, $u[2]$, and $u[3]$.
Your expressions can have other relevant variables (e.g. A, b etc) and mathematical operators.

Answer:

From Equation (1), we get (by substituting $n = 1$):

$$\mathbf{x}[2] = \mathbf{A}\mathbf{x}[1] + \mathbf{b}\,u[1]$$

By substituting $\mathbf{x}[1]$ from Equation (2), we get

$$\begin{aligned}\mathbf{x}[2] &= \mathbf{A}\mathbf{x}[1] + \mathbf{b}\,u[1] \\ &= \mathbf{A}\left(\mathbf{A}\mathbf{x}[0] + \mathbf{b}\,u[0]\right) + \mathbf{b}\,u[1] \\ &= \mathbf{A}^2\mathbf{x}[0] + \mathbf{A}\mathbf{b}\,u[0] + \mathbf{b}\,u[1]\end{aligned} \tag{3}$$

From Equation (1), we get (by substituting $n = 2$):

$$\mathbf{x}[3] = \mathbf{A}\mathbf{x}[2] + \mathbf{b}\,u[2]$$

By substituting $\mathbf{x}[2]$ from Equation (3), we get

$$\begin{aligned}\mathbf{x}[3] &= \mathbf{A}\mathbf{x}[2] + \mathbf{b}\,u[2] \\ &= \mathbf{A}\left(\mathbf{A}^2\mathbf{x}[0] + \mathbf{A}\mathbf{b}\,u[0] + \mathbf{b}\,u[1]\right) + \mathbf{b}\,u[2] \\ &= \mathbf{A}^3\mathbf{x}[0] + \mathbf{A}^2\mathbf{b}\,u[0] + \mathbf{A}\mathbf{b}\,u[1] + \mathbf{b}\,u[2]\end{aligned} \tag{4}$$

From Equation (1), we get (by substituting $n = 3$):

$$\mathbf{x}[4] = \mathbf{A}\mathbf{x}[3] + \mathbf{b}\,u[3]$$

By substituting $\mathbf{x}[3]$ from Equation (4), we get

$$\begin{aligned}\mathbf{x}[4] &= \mathbf{A}\mathbf{x}[3] + \mathbf{b}\,u[3] \\ &= \mathbf{A}\left(\mathbf{A}^3\mathbf{x}[0] + \mathbf{A}^2\mathbf{b}\,u[0] + \mathbf{A}\mathbf{b}\,u[1] + \mathbf{b}\,u[2]\right) + \mathbf{b}\,u[3] \\ &= \mathbf{A}^4\mathbf{x}[0] + \mathbf{A}^3\mathbf{b}\,u[0] + \mathbf{A}^2\mathbf{b}\,u[1] + \mathbf{A}\mathbf{b}\,u[2] + \mathbf{b}\,u[3] \end{aligned} \tag{5}$$

(d) Now, generalize the pattern you saw in the earlier part to write an expression for $\mathbf{x}[N]$ where $N \in \mathbb{Z}^+$ in terms of $\mathbf{x}[0]$ and the inputs from $u[0], \ldots, u[N-1]$. **Your expression can have other relevant variables (e.g. A, b etc) and mathematical operators.**

Answer:

Use the same procedure as above for N steps. You will obtain the following expression:

$$\mathbf{x}[N] = \mathbf{A}^N\mathbf{x}[0] + \mathbf{A}^{N-1}\mathbf{b}\,u[0] + \ldots + \mathbf{A}\mathbf{b}\,u[N-2] + \mathbf{b}\,u[N-1] \tag{6}$$

You might also use the compact expression:

$$\mathbf{x}[N] = \mathbf{A}^N\mathbf{x}[0] + \sum_{i=0}^{N-1} \mathbf{A}^i\mathbf{b}\,u[N-i-1] \tag{7}$$

Note that $\mathbf{A}^0$ is the identity matrix.

Also, $\sum_{i=0}^{N-1} \mathbf{A}^i\mathbf{b}\,u[N-i-1] = \sum_{i=0}^{N-1} \mathbf{A}^{N-i-1}\mathbf{b}\,u[i]$. So, either of the two expressions can be used in Equation 7.

As a sanity check, plug the values $N = 1, 2, 3$, and 4 to obtain Equations (2), (3), (4), and (5), respectively.

For the next two parts of the problem, you are given the matrix $\mathbf{A}$ and the vector $\mathbf{b}$:

$$\mathbf{A} = \begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \\ 2 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Assume the cart-pole starts in an initial state $\mathbf{x}[0] = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$, and you want to reach the desired state $\mathbf{x}_f = \mathbf{0}$ using the control inputs $u[0], u[1], \ldots$ etc. The state vector $\mathbf{x}_f = \mathbf{0}$ corresponds to the cart-pole (or segway) being upright and stopped at the origin. **Reaching $\mathbf{x}_f = \mathbf{0}$ in N steps means that, given that we start at $\mathbf{x}[0]$, we can find control inputs ($u[0], u[1], \ldots$ etc), such that we get $\mathbf{x}[N]$ (i.e. state vector at Nth time step) equal to $\mathbf{x}_f = \mathbf{0}$.**

(e) Can you reach $\mathbf{x}_f$ in *two* time steps? If so, find the series of inputs $\mathbf{u}$ that enables us to reach $\mathbf{x}_f$.(*Hint: Express* $\mathbf{x}[2] - \boldsymbol{A}^2\mathbf{x}[0]$ *in terms of the inputs* $u[0]$ *and* $u[1]$. *Then determine if the system of equations can be solved to obtain* $u[0]$ *and* $u[1]$. *If we obtain valid solutions for* $u[0]$ *and* $u[1]$, *then we can say we will reach* $\mathbf{x}_f$ *in two time steps.*)

Answer:

No.

From Equation (3), we know that $\mathbf{A}^2\mathbf{x}[0] + \mathbf{A}\mathbf{b}\,u[0] + \mathbf{b}\,u[1] = \mathbf{x}[2]$ which is equivalent to $\mathbf{A}\mathbf{b}\,u[0] + \mathbf{b}\,u[1] = \mathbf{x}[2] - \mathbf{A}^2\mathbf{x}[0]$.

This means that in order to reach any state $\mathbf{x}_f$ in two time steps (that is, $\mathbf{x}[2] = \mathbf{x}_f$), we have to solve the following system of linear equations:

$$\mathbf{A}\mathbf{b}\,u[0] + \mathbf{b}\,u[1] = \mathbf{x}_f - \mathbf{A}^2\mathbf{x}[0],$$

where $u[0]$ and $u[1]$ are the unknowns.

Since in our case we want to reach $\mathbf{x}_f = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, the system of linear equations simplifies to

$$\mathbf{A}\mathbf{b}\,u[0] + \mathbf{b}\,u[1] = -\mathbf{A}^2\mathbf{x}[0].$$

In matrix form, this system of linear equations is

$$\begin{bmatrix} | & | \\ \mathbf{A}\mathbf{b} & \mathbf{b} \\ | & | \end{bmatrix} \begin{bmatrix} u[0] \\ u[1] \end{bmatrix} = -\mathbf{A}^2\mathbf{x}[0],$$

which yields the following augmented matrix:

$$\left[\begin{array}{cc|c} | & | & | \\ \mathbf{A}\mathbf{b} & \mathbf{b} & -\mathbf{A}^2\mathbf{x}[0] \\ | & | & | \end{array}\right].$$

By plugging in the values of $\mathbf{A}$, $\mathbf{b}$, and $\mathbf{x}[0]$, we get the following augmented matrix:

$$\left[\begin{array}{cc|c} 0 & 0 & -4 \\ 0 & 0 & 0 \\ 2 & 0 & -4 \\ 0 & 1 & -4 \end{array}\right].$$

which means that the system is inconsistent (due to the first row) and that there are no solutions for $u[0]$ and $u[1]$. It is fine if you did not row reduce all the way to the upper triangular form as long as you showed that the system of equations is inconsistent.

(f) Can you reach $\mathbf{x}_f$ in *three* time steps? If so, find the series of inputs $\mathbf{u}$ that enables us to reach $\mathbf{x}_f$.

Answer:

Yes.

Similar to the previous part, from Equation (4), we know that $\mathbf{A}^3\mathbf{x}[0] + \mathbf{A}^2\mathbf{b}\,u[0] + \mathbf{A}\mathbf{b}\,u[1] + \mathbf{b}\,u[2] = \mathbf{x}[3]$, which is equivalent to $\mathbf{A}^2\mathbf{b}\,u[0] + \mathbf{A}\mathbf{b}\,u[1] + \mathbf{b}\,u[2] = \mathbf{x}[3] - \mathbf{A}^3\mathbf{x}[0]$.

This means that in order to reach any state $\mathbf{x}_f$ in three time steps (that is, $\mathbf{x}[3] = \mathbf{x}_f$), we have to solve the following system of linear equations:

$$\mathbf{A}^2\mathbf{b}\,u[0] + \mathbf{A}\mathbf{b}\,u[1] + \mathbf{b}\,u[2] = \mathbf{x}_f - \mathbf{A}^3\mathbf{x}[0],$$

where $u[0], u[1]$, and $u[2]$ are the unknowns.

Since in our case we want to reach $\mathbf{x}_f = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, the system of linear equations simplifies to

$$\mathbf{A}^2\mathbf{b}\,u[0] + \mathbf{A}\mathbf{b}\,u[1] + \mathbf{b}\,u[2] = -\mathbf{A}^3\mathbf{x}[0].$$

In matrix form, this system of linear equations is

$$\begin{bmatrix} | & | & | \\ \mathbf{A}^2\mathbf{b} & \mathbf{A}\mathbf{b} & \mathbf{b} \\ | & | & | \end{bmatrix} \begin{bmatrix} u[0] \\ u[1] \\ u[2] \end{bmatrix} = -\mathbf{A}^3\mathbf{x}[0],$$

which yields the following augmented matrix:

$$\left[\begin{array}{ccc|c} | & | & | & | \\ \mathbf{A}^2\mathbf{b} & \mathbf{A}\mathbf{b} & \mathbf{b} & -\mathbf{A}^3\mathbf{x}[0] \\ | & | & | & | \end{array}\right].$$

By plugging in the values of $\mathbf{A}$, $\mathbf{b}$, and $\mathbf{x}[0]$, we get the following augmented matrix:

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 4 & 0 & 0 & -8 \\ 0 & 2 & 0 & -8 \\ 0 & 0 & 1 & -8 \end{array}\right].$$

Applying Gaussian elimination, we get the upper triangular form to be

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & -8 \\ 0 & 0 & 0 & 0 \end{array}\right],$$

This indicates that there exists a solution to the system of equations, which is

$$\begin{bmatrix} u[0] \\ u[1] \\ u[2] \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \\ -8 \end{bmatrix},$$

(g) Let's try to generalize the idea in part (a). Express the states you can reach in N timesteps as a span of some vectors plus a vector offset. (Hint: Consider the direction that each control input $u[0], \ldots, u[N-1]$ can move $\mathbf{x}[N]$ by.)

Answer: In N timesteps, the control inputs $u[0], \ldots, u[N-1]$ allow you to move the state $\mathbf{x}[N]$ by any vector in

$$\text{span}\left\{\mathbf{b}, \mathbf{A}\mathbf{b}, \mathbf{A}^2\mathbf{b}, \ldots, \mathbf{A}^{N-1}\mathbf{b}\right\}.$$

You will also have an offset of $\mathbf{A}^N\mathbf{x}[0]$, which captures the zero input system behavior. Taking this into consideration you can reach all states that are in

$$\text{span}\left\{\mathbf{b}, \mathbf{A}\mathbf{b}, \mathbf{A}^2\mathbf{b}, \ldots, \mathbf{A}^{N-1}\mathbf{b}\right\} + \mathbf{A}^N\mathbf{x}[0].$$